FINAL YEAR PROJECT

Clinic Queue & Patient Flow Management System

A smart, web-based solution for university clinic patient registration, prioritization, and real-time queue management.

Presented by · Computer Science Department · 2025

THE PROBLEM

University clinics are overwhelmed.

Long queues, missed emergencies, and paper-based systems create a broken patient experience — costing time and, in critical cases, lives.

Overcrowding & Long Waits

No visibility into queue length or wait time

Missed Emergencies

Critical patients not triaged fast enough

Paper-Based Inefficiency

Manual records cause errors and slow throughput

OUR SOLUTION

A web-based smart queue system

Built for real university clinic conditions — no app install required, no hardware needed.

Smart Registration

Patients register via browser with symptom input and instant queue assignment

Priority Engine

Automated triage ranks patients by emergency level, symptoms, and age

Live Dashboard

Staff get a real-time queue board and reporting tools at a glance

KEY FEATURES

Smart Registration

Web-based patient intake with symptom capture and real-time queue placement

Emergency Detection

Symptom-based triage algorithm with age factor for automatic priority scoring

Queue Display Board

Live public screen showing current queue state — zero staff intervention needed

Reporting Dashboard

Admin analytics: daily volumes, wait times, peak hours, and case history

PRIORITY DETECTION

How emergencies are caught instantly

01

Symptom Input

Patient selects symptoms during registration — chest pain, difficulty breathing, etc.

02

Logic Scoring

Each symptom carries a severity weight; combined score determines urgency tier

03

Age Prioritization

Elderly (60+) and infant patients automatically receive elevated priority tier

04

Queue Positioning

High-priority patients jump the queue; staff dashboard flags them in red

A clean, layered architecture for reliability and speed

Frontend Interface

HTML5 · CSS3 · JavaScript

Queue Engine

Priority logic · State manager

Priority Detection

Symptom scoring · Age factors

Dashboard & Reports

Analytics · Admin view

Each layer is independently testable — the system degrades gracefully if any component is unavailable.

TECHNOLOGY STACK

Built lean — runs everywhere

HTML5

Semantic structure & accessibility

CSS3

Responsive layouts & animations

JavaScript

Queue logic, DOM control & state

No Backend

Zero server dependency for v1.0

No Install

Runs in any modern browser

Scalable

Designed for DB integration in v2.0

IMPACT & BENEFITS

60%

Faster triage

Automated priority detection vs manual assessment

100%

Emergency capture rate

No patient skipped — every admission is scored

Zero

Hardware required

Runs on any browser — phones, tablets, or PCs

FUTURE ROADMAP

Where we go from here

Phase 2

Database Integration

Persistent MySQL/PostgreSQL backend with patient history and analytics export

Phase 3

Cloud Deployment

Host on AWS or Firebase for multi-clinic, institution-wide access

Phase 4

SMS Notifications

Patients receive queue updates via SMS — no need to wait inside the clinic

Phase 5

Mobile App (Flutter)

Native mobile app version for patients and staff with offline capability

Thank You

Clinic Queue & Patient Flow Management System

Questions & Discussion Welcome

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